

Marketing Funnel & Conversion Performance Analysis

Objective

The objective of this project is to analyze marketing and e-commerce order data to identify revenue trends, customer purchasing behavior, payment preferences, and order performance. The analysis helps understand conversion patterns and provides insights that can improve business decisions and customer experience.

Dataset

Dataset Used: Olist Brazilian E-commerce Dataset

The dataset contains information about customer orders, payments, products, and customers. Multiple CSV files were merged to perform a complete business analysis.

Tools Used

- Python
- Pandas
- Matplotlib
- Power BI
- VS Code

Data Preparation

The following preprocessing steps were performed:

- Imported multiple CSV datasets.
- Merged order, customer, payment, and order item datasets.
- Removed duplicate records.
- Removed missing values.
- Exported the cleaned dataset for Power BI visualization.

Dashboard Overview

The Power BI dashboard includes:

- Total Revenue
- Total Orders
- Total Customers
- Average Payment Value
- Revenue by Payment Type
- Orders by Status
- Monthly Revenue Trend
- Revenue by Customer State
- Payment Type Distribution
- Interactive slicers for Payment Type, Order Status, and Customer State.

Key Insights

- The dashboard provides a clear overview of overall business performance.
- Revenue is generated through multiple payment methods, with some methods contributing more than others.
- Most orders are successfully delivered, while a smaller percentage are canceled or unavailable.
- Revenue trends vary over time, helping identify peak sales periods.
- Certain customer states contribute significantly more revenue than others.

Recommendations

- Promote the most frequently used payment methods to improve customer convenience.
- Reduce canceled and delayed orders by improving logistics and inventory management.

- Focus marketing campaigns on high-performing regions to maximize sales.
- Monitor monthly revenue trends to plan seasonal promotions.
- Continue tracking customer purchasing behavior to improve retention and conversion rates.

Conclusion

This project successfully analyzed marketing and e-commerce performance using Python and Power BI. The dashboard provides valuable insights into revenue, payment methods, customer behavior, and order performance. These insights can support data-driven business decisions and help improve overall conversion performance.

Dashboard Screenshot

